

Software Requirements Specification (SRS)

Project: CSE Permission Tracking System

Department of Computer Science & Engineering
Aditya University

Version: 1.0

1 Introduction

1.1 Purpose

This document specifies the software requirements for the CSE Permission Tracking System. The system replaces the traditional, paper-based permission and leave letter process with a secure, digital workflow. It is designed to eliminate physical forgery, streamline faculty approvals, and provide real-time oversight for departmental leadership.

1.2 Scope

The system is a web-based application utilizing Microsoft Outlook Single Sign-On (SSO) for authentication. Students will submit digital requests with mandatory proof, which will trigger automated email notifications to faculty. Faculty will review and manage these requests, while the Head of Department (HOD) will have access to a real-time, analytical dashboard with report-printing capabilities.

2 Overall Description

2.1 User Roles

Role	Responsibilities & Access Levels
Student	Authenticates via Outlook SSO. Submits permission requests containing name, roll number, phone number, reason, and digital proof (image/PDF).
Lecturer / Faculty	Receives email alerts for new requests. Logs in to review attached proof and securely approves or rejects student requests.
HOD	Accesses an administrative dashboard displaying real-time metrics, approval percentages, and a printable daily roster of permitted students.

2.2 Operating Environment

- **Hosting:** Aditya University Internal Servers (on-premise deployment).
- **Storage:** Local server directory mapping for secure, internal document storage (zero external cloud dependencies).
- **Client Interface:** Modern web browsers (Chrome, Edge, Safari) across both desktop and mobile devices.

2.3 Technology Stack

- **Frontend:** HTML5, CSS3, Bootstrap 5 (Responsive UI), Chart.js (Data Visualization).
- **Backend:** Python 3 (Flask Framework).
- **Authentication:** Microsoft Authentication Library (MSAL) for Azure Active Directory / Outlook SSO.
- **Database:** MySQL.
- **Email Services:** Flask-Mail (routing via Aditya University Outlook SMTP).

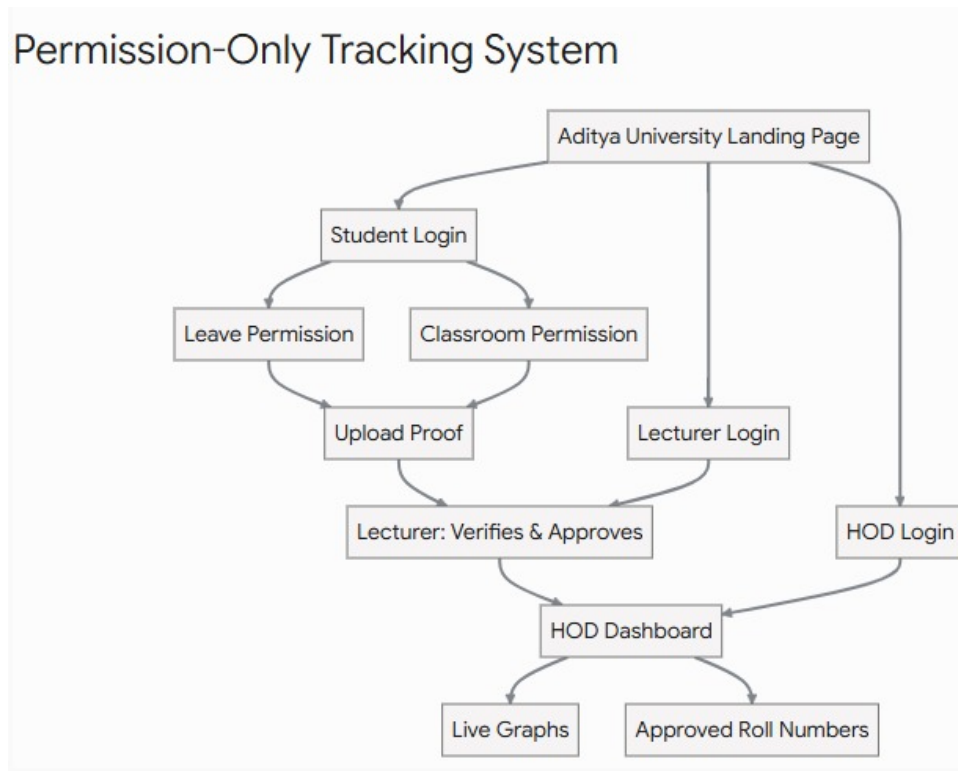


Figure 1: Architecture of Permission Tracking System

3 System Features (Functional Requirements)

3.1 Microsoft Outlook Authentication (SSO)

Description: The system must restrict access exclusively to active Aditya University accounts.

Requirements:

- Users click a "Sign in with Aditya Outlook" button.
- Authentication is handled securely via Microsoft Azure without storing user passwords in the local MySQL database.
- The system automatically captures the user's verified college email to determine their role based on the database mapping.

3.2 Student Permission Submission Module

Description: The secure interface for students to request permissions.

Requirements:

- The form must mandate the following fields: Full Name, Roll Number, Phone Number, Reason for Permission, and Proof Document.
- Proof documents must be strictly validated for file type (e.g., .jpg, .png, .pdf) and securely saved to the designated Aditya server storage.
- The database will store the file path reference, not the binary file itself, to optimize database performance.

3.3 Faculty Notification & Action Workflow

Description: The automated routing and approval system for teaching staff.

Requirements:

- Upon successful student submission, an automated email must be immediately dispatched to the relevant faculty member's Outlook inbox.
- The email must contain the student's name, roll number, and a direct link to the pending request.
- Faculty must be able to view the uploaded proof directly within the portal and execute an "Approve" or "Reject" action, updating the database status instantly.

3.4 HOD Dashboard & Analytics

Description: The executive overview for departmental leadership.

Requirements:

- The dashboard must display dynamic visual charts (via Chart.js) showing the percentage of approved vs. rejected requests and categorizing reasons for leave.
- The dashboard must generate a live, filterable roster of all approved permissions for the current day.

3.5 Report Generation & Printing

Description: Physical record generation capability.

Requirements:

- A dedicated "Print Report" function must be available on the HOD and Faculty dashboards.
- The print layout must automatically strip out navigational elements and UI buttons, formatting the data as a clean, official PDF/paper document suitable for departmental archiving.

4 Non-Functional Requirements

4.1 Security & Privacy

- All student proof documents (medical records, letters) must remain strictly on Aditya University's internal servers to comply with institutional data privacy standards.
- Files must be sanitized upon upload (e.g., using `secure_filename()`) to prevent directory traversal attacks.

4.2 Usability

- The frontend interface must be fully responsive, ensuring the UI (especially the student request form) is easily navigable and functional on mobile devices.

4.3 Reliability

- The application must gracefully handle SMTP timeout errors or Microsoft SSO temporary outages, providing clear fallback instructions to the user rather than crashing the web server.

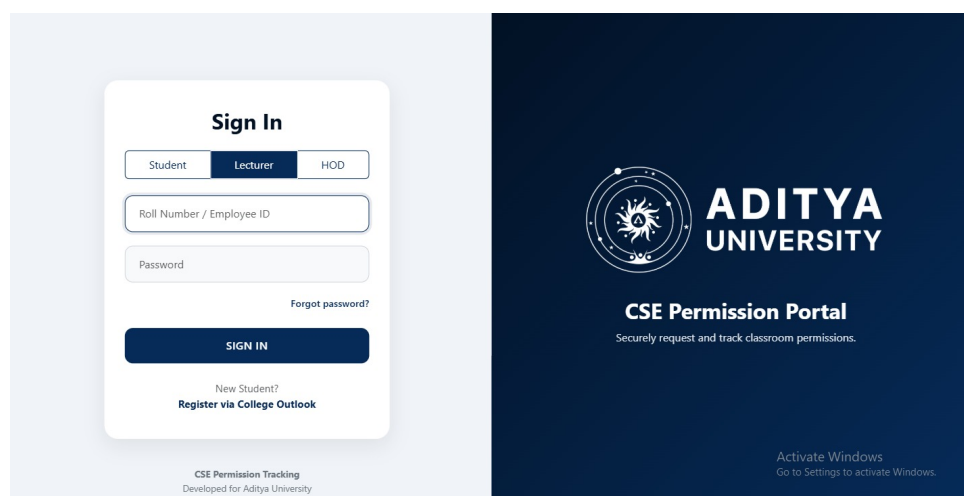


Figure 2: Aditya University CSE Permission Portal